

The urban exposome of Santiago

An open-data, reproducible demonstration of commune-level environmental and social exposure layers

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1. Purpose

The *exposome* is the set of environmental and social exposures that accumulate over a person's life and influence brain health and the risk of cognitive decline and dementia (Livingston et al., 2020). At BrainLat, these exposures usually have to be reconstructed from where each participant lives. This document is a concrete example of how I would build them, and of the data and tools that can keep refining them. Using only **open data**, I derive more than twenty exposome layers for the **52 communes of the Metropolitan Region of Santiago** — air quality, industrial heavy metals, green space, artificial light at night, urban heat, precipitation, wildfire, socioeconomic level, healthcare access, and others — integrated into a single master table (52 communes, 261 variables, no gaps). Figure 1 shows five representative layers.

Each layer is a self-contained, reproducible notebook and needs no API keys. A static open-data snapshot still misses a great deal, and that is where stronger sources matter. **Time-domain satellite data** (Sentinel-2 NDVI series, Landsat) and **computer vision** recover exposures that open maps leave out. Green space is the clearest case: directly from aerial imagery I detect the private gardens and street trees that OpenStreetMap never registers (Fig. 2). Reading satellite archives as time series, to follow change and flag gaps, is the same work I did in time-domain astronomy.

Urban exposome of Santiago — five reproducible commune-level layers

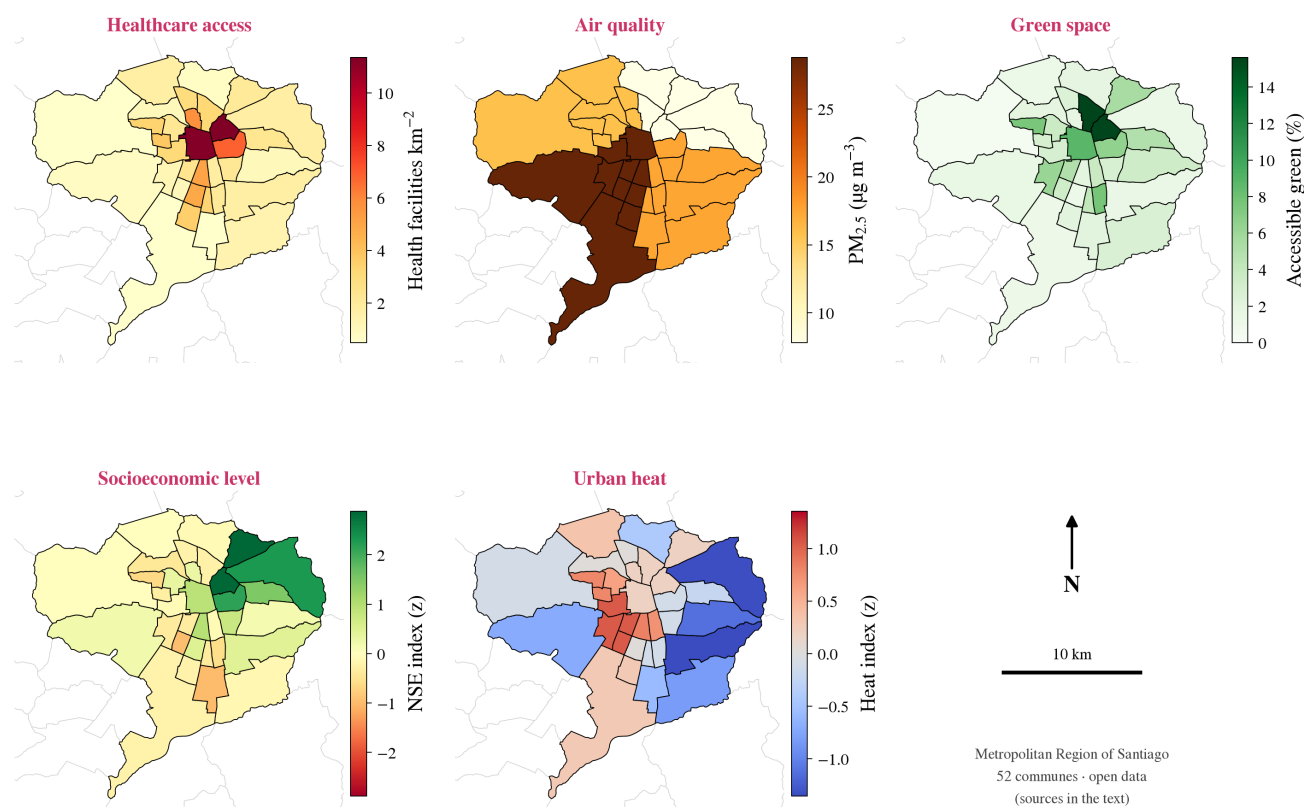


Figure 1: Five commune-level exposome layers for the Metropolitan Region of Santiago, each harmonised to the same 52 communes. The colour scale of every panel and how it is computed are described in Section 2. Diverging scales (socioeconomic level, urban heat) are standardised indices centred at the regional mean; sequential scales are physical magnitudes. The same east-west gradient recurs across panels: the high-income, leafy, cooler east versus the more polluted, hotter, lower-income centre and south-west.

2. Key exposome layers: indicator, computation and source

For each layer I state *what the colour scale represents, how it is computed, and the data source*.

Healthcare access

Colour scale: density of health facilities (establishments per km²). Health amenities (hospital, clinic, pharmacy, doctors, dentist) are queried from OpenStreetMap, assigned to communes by spatial join, counted, and divided by the commune area in a metric projection (EPSG:32719); the notebook also derives mean / median / P90 distance to the nearest facility on an intra-commune grid. **Source:** OpenStreetMap via OSMnx (Boeing, 2017; OpenStreetMap contributors, 2024).

Air quality

Colour scale: annual-mean fine particulate matter PM_{2.5} (μg m⁻³). The CAMS atmospheric reanalysis is sampled on a ~11 km grid over the region (hourly, full year), averaged to an annual mean and aggregated to communes; values are benchmarked against the WHO 2021 annual guideline of 5 μg m⁻³ (every commune exceeds it). **Source:** Open-Meteo Air-Quality API (CAMS), cross-checked against the official SINCA ground stations (Open-Meteo, 2024; Copernicus Atmosphere Monitoring Service (CAMS), 2024; World Health Organization, 2021; Ministerio del Medio Ambiente de Chile, 2024).

Industrial heavy metals (RETC)

Colour scale: log(Pb + 1) from industrial point-source air emissions [kg yr⁻¹, log scale]. Lead, arsenic and mercury from industrial facilities declared to the Chilean national pollutant release register (RETC, Ministry of the Environment). Annual facility-level emissions are downloaded for 2015–2022 (CSV + XLSX, no API key), unit-converted to kg/yr, assigned to communes by point-in-polygon spatial join, and averaged across years. Lead is the single best-documented neurotoxin in the Lancet Commission on Dementia 2024: exposure reduces hippocampal volume and accelerates cognitive decline dose-responsively. The most striking finding is that Tiltit, a semi-rural northern commune, concentrates ~99.6% of all industrial Pb emissions in the Metropolitan Region (~10 629 kg/yr), attributable to a known industrial complex; the Spearman correlation between log(Pb) and PM_{2.5} is $\rho \approx -0.15$ ($p = 0.30$, n.s.), confirming that industrial point-source neurotoxins and combustion-driven particulate pollution are *independent* exposure pathways. **Source:** RETC MMA (datosretc.mma.gob.cl), CC-BY, 2015–2022.

Green space

Colour scale: share of the commune covered by accessible green space (%). Parks, gardens, squares and recreation grounds are extracted from OpenStreetMap, dissolved into a single geometry (to avoid double counting), intersected with each commune, and divided by the commune area. Because this layer measures accessible, mapped green areas rather than total vegetation, Section 4 uses aerial-imagery computer vision to quantify the residual greenery that OpenStreetMap leaves out. **Source:** OpenStreetMap (OpenStreetMap contributors, 2024).

Socioeconomic level

Colour scale: composite socioeconomic index (standardised, regional mean = 0). Per-commune household income and years of schooling (CASEN 2022) and the official small-area poverty estimate are each converted to z -scores; the index is the mean of income (+), schooling (+) and poverty (−). **Source:** Ministry of Social Development (CASEN, SAE poverty) via open data (Ministerio de Desarrollo Social y Familia, 2023; Olea Abarca, 2024).

Urban heat

Colour scale: heat-exposure index (standardised, regional mean = 0). Daily 2024 temperatures are summarised per commune into summer mean $T_{\max}$, $T_{\max}$ P95, days $\geq 30/35$ °C and tropical nights ($T_{\min} \geq 20$ °C); these are z -scored and averaged. To avoid biasing large pre-Andean communes with uninhabited high-mountain pixels, the residential estimate uses only each commune's low-elevation (valley) band, identified with point elevations. **Source:** Open-Meteo Historical Weather (ERA5-based reanalysis) (Open-Meteo, 2024).

3. Integration and a first finding

All five layers share the same 52 communes, so the master table supports immediate cross-analysis. A simple integration already yields a non-trivial result: socioeconomic level correlates *significantly* with access to green space (Spearman $\rho \approx +0.44$, $p < 0.001$) and to health services ($\rho \approx +0.59$, $p < 0.001$), but *not* with PM_{2.5} ($\rho \approx +0.09$, $p \approx 0.54$, n.s.) or NO₂ ($p \approx 0.74$, n.s.); all are Spearman correlations with two-sided p -values over the $n=52$ communes (the critical $|\rho|$ for $p < 0.05$ is ≈ 0.27). In Santiago, environmental inequity therefore operates mainly through unequal access to green space and services, while air pollution is governed more by the basin's topography and thermal inversion than by income — exactly the kind of harmonised, multi-layer inference that exposome research at BrainLat requires.

Layer vs. socioeconomic level	ρ	p	
Green space (% accessible)	+0.44	< 0.001	***
Healthcare facility density	+0.59	< 0.001	***
Urban heat index	-0.14	0.34	n.s.
PM _{2.5} (annual mean)	+0.09	0.54	n.s.
NO ₂ (annual mean)	+0.05	0.74	n.s.
Industrial Pb log(kg/yr)	+0.22	0.11	n.s.

Spearman rank correlations of each exposome layer against the composite socioeconomic index ($n = 52$ communes; two-sided p -values; $|\rho| \gtrsim 0.27$ is significant at $p < 0.05$). *** $p < 0.001$; n.s. = not significant. Green space and healthcare show strong socioeconomic gradients; air quality and industrial heavy metals do not — Pb emissions in Santiago are driven by a single industrial complex in the semi-rural commune of Tiltit, not by deprivation geography.

4. From an open baseline to satellite, time-domain exposure

This demonstration is deliberately a **reproducible baseline**, not a finished product. Its most useful property is that it makes *the incompleteness of current open samples visible* — which is precisely where satellite Earth-observation data, and especially **time-domain** satellite data, add value:

- **Green space.** OpenStreetMap maps *public* parks but misses private gardens, street trees and seasonal vegetation, and under-represents affluent low-density areas. A computer-vision baseline on public aerial imagery — the Excess Green index (Woebbecke et al., 1995) thresholded with Otsu’s method (Otsu, 1979), with a green-dominance test and morphological filtering to suppress false positives — makes this concrete (Fig. 2). **It detects substantially more vegetation than the open layer — of order 6–13× in leafy residential sectors — because ~94% of the detected greenery (private gardens and street trees) lies outside any OSM-mapped park.** Multispectral Sentinel-2 NDVI (Copernicus / ESA, 2024) extends this to full regional coverage and seasonality, and Landsat back decades.
- **Air quality and heat.** CAMS is a coarse (~11 km) model field. Sentinel-5P (NO₂, aerosols) and Landsat/Sentinel-3 land-surface temperature resolve intra-urban gradients and the neighbourhood-scale urban heat island that a commune average smooths out (Copernicus / ESA, 2024).
- **Data completeness as a time-domain problem.** Treating the satellite archive as a time series lets us *detect gaps, outliers and change* — new parks, construction, deforestation — and flag communes whose OpenStreetMap or model layers are stale or under-sampled. Earth observation thus becomes a quality-control layer over the open baseline, rather than a one-off snapshot.

This is a direct transfer of my background in **time-domain astronomy** (multi-epoch light curves from ATLAS, ZTF and the Vera C. Rubin Observatory): multi-epoch calibration, gap and outlier handling, anomaly/change detection, and large heterogeneous data pipelines map one-to-one onto satellite revisit time series for exposome mapping. The computer-vision example above (vegetation segmentation on imagery) is the same competence applied to the spatial domain.

5. Reproducibility

All code — one notebook or script per factor, the master integration script (scripts/build_master_exposome.py), a conda environment.yml, and the scripts that generate every figure in this document — is openly available at <https://github.com/Bastian963/demo-exposome-santiago>. Every layer is harmonised to the 52 official communes of the Metropolitan Region, computed in a metric projection (EPSG:32719) and exported as both a table (CSV) and a GIS layer (GeoJSON). All inputs are open; most layers require no authentication (RETC, OpenStreetMap, CASEN, INE Census 2017); satellite layers use Google Earth Engine as an optional extension. Each script caches raw responses, so re-runs are offline and deterministic, and the master builder validates a strict one-to-one join (52 rows, 261 variables, no missing values) before writing.

Repository layout.

notebooks/santiago*.ipynb the factor notebooks and the optional computer-vision example.

scripts/run*.py config-driven CLI scripts (one per layer); prefer over notebooks for new layers.

data/processed/*.{csv,geojson,json} per-factor commune indicators as a table, a GIS layer and machine-readable metadata.

figures/*_santiago*.png , maps/*_santiago.html publication-quality static maps and interactive web maps per factor.

scripts/build_master_exposome.py joins all layers and writes the master CSV, GeoJSON and metadata (52 × 261).

docs/*_methodology.md per-layer methodology, data sources, limitations and key findings.

environment.yml , README.md reproducible conda environment and project documentation.

Data sources and references

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- Woebbecke, D. M. et al. (1995). "Color indices for weed identification under various soil, residue, and lighting conditions". In: *Transactions of the ASAE* 38.1, pp. 259–269. DOI: [10.13031/2013.27838](https://doi.org/10.13031/2013.27838).
- World Health Organization (2021). *WHO global air quality guidelines: particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. <https://www.who.int/publications/i/item/9789240034228>.



Figure 2: Worked example — detecting green space with computer vision. A transparent pipeline — Excess Green index (Woebecke et al., 1995) thresholded with Otsu's method (Otsu, 1979), plus a green-dominance test and morphological cleaning to suppress false positives — outlines per-pixel vegetation (lime) on public aerial imagery. Overlaying the OpenStreetMap green polygons (cyan) shows that **~94% of the detected vegetation in the residential sector lies outside any OSM-mapped park**: it is street trees and private gardens, absent from the OSM accessible-green layer that underlies the green file (Section 2). The contours track tree canopy and gardens, not roofs, streets or shadows. The same approach scales to CNN / foundation-model segmenters (SAM, DeepLabv3+) and to multispectral Sentinel-2 NDVI. *Imagery* © Esri, Maxar, Earthstar Geographics.